

Predicting late deliveries before they happen

Same model, same scores. The only thing that changed was where I drew the line.

Share of late orders caught on unseen future months

Default cut (0.5)



7.1%

Chosen cut (0.28)



59.7%

610

of 1,021 late orders
caught in the test months

2.0×

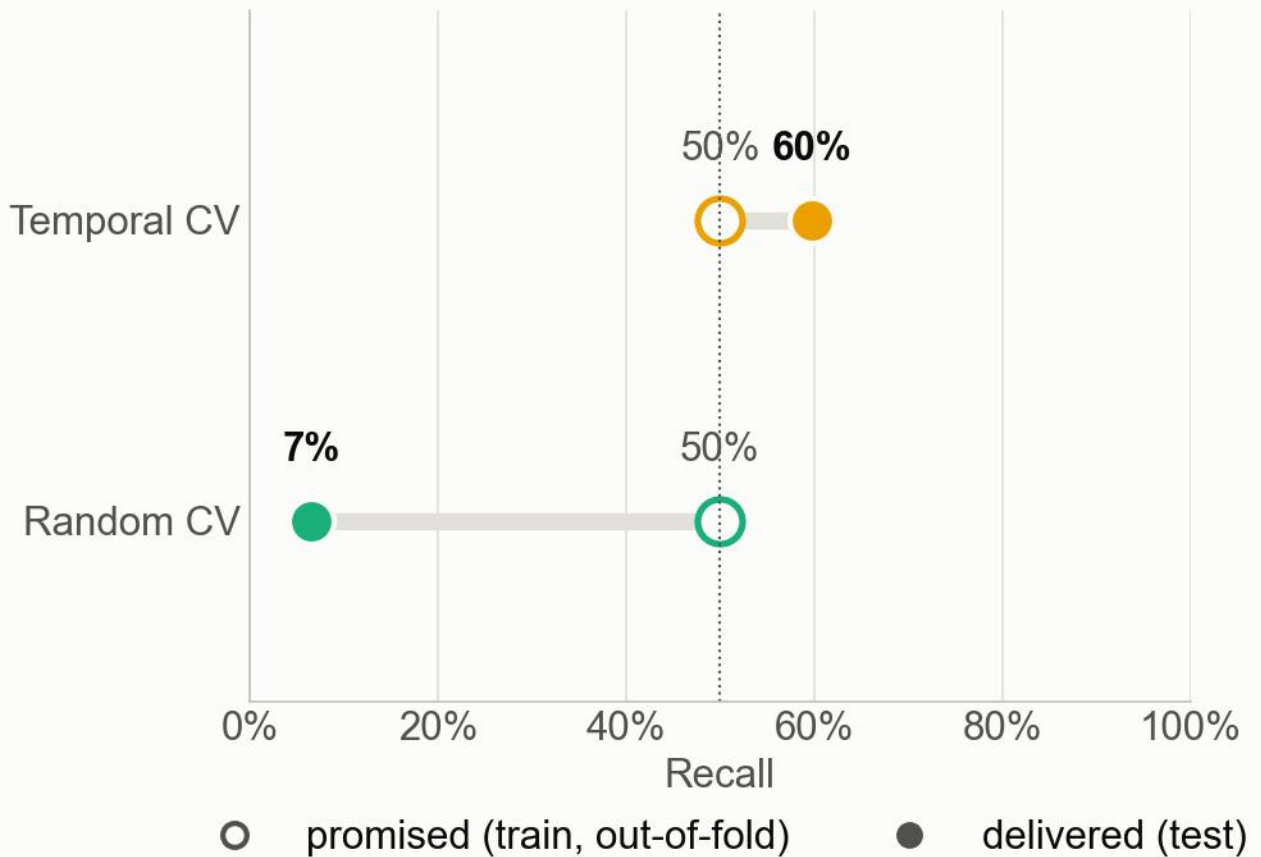
the base late rate among
flagged orders

30%

of orders flagged
for proactive action

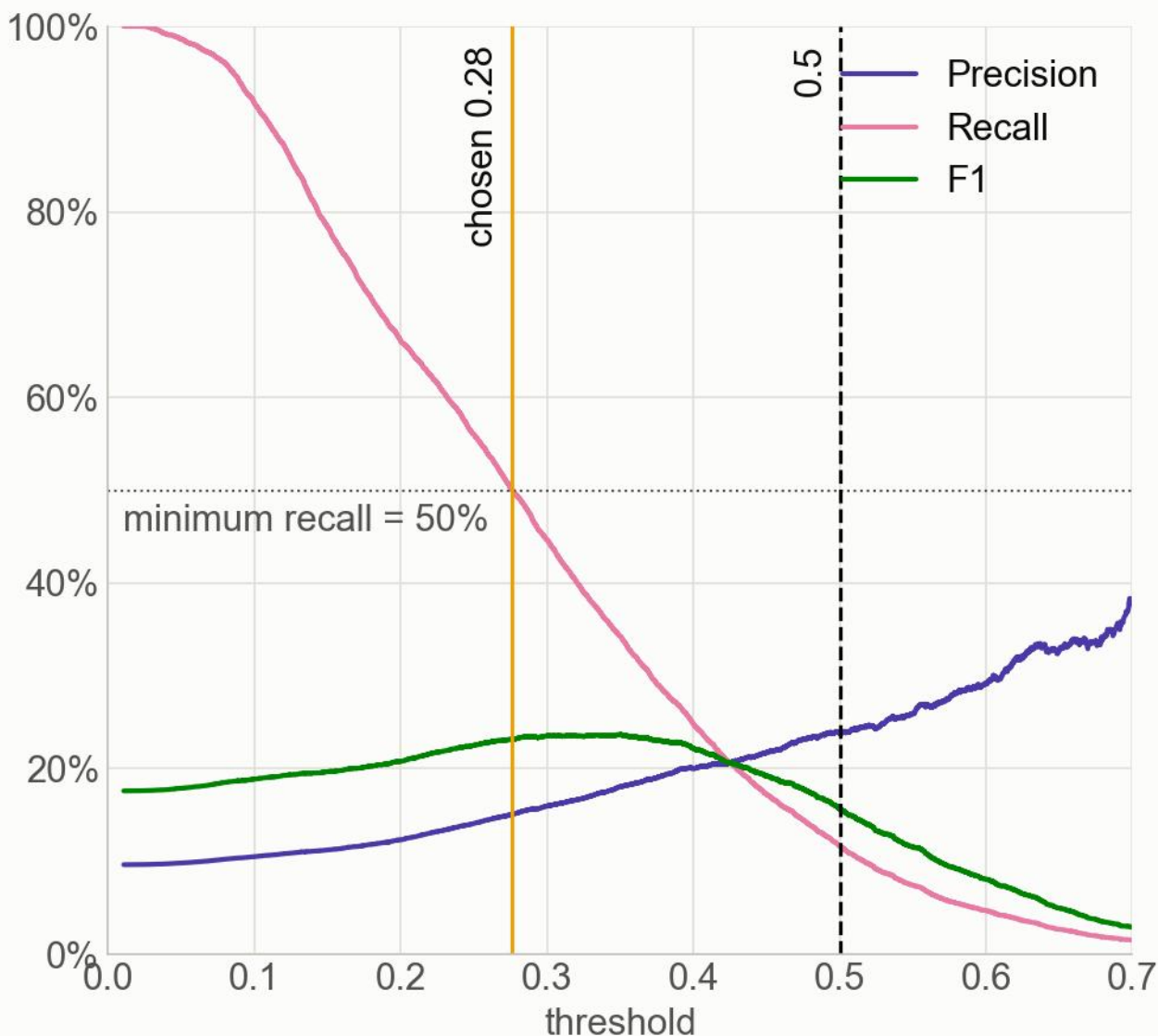
A threshold picked on random CV breaks in production

Both cuts promised 50% recall on the training data.
Only the time-aware one kept the promise.



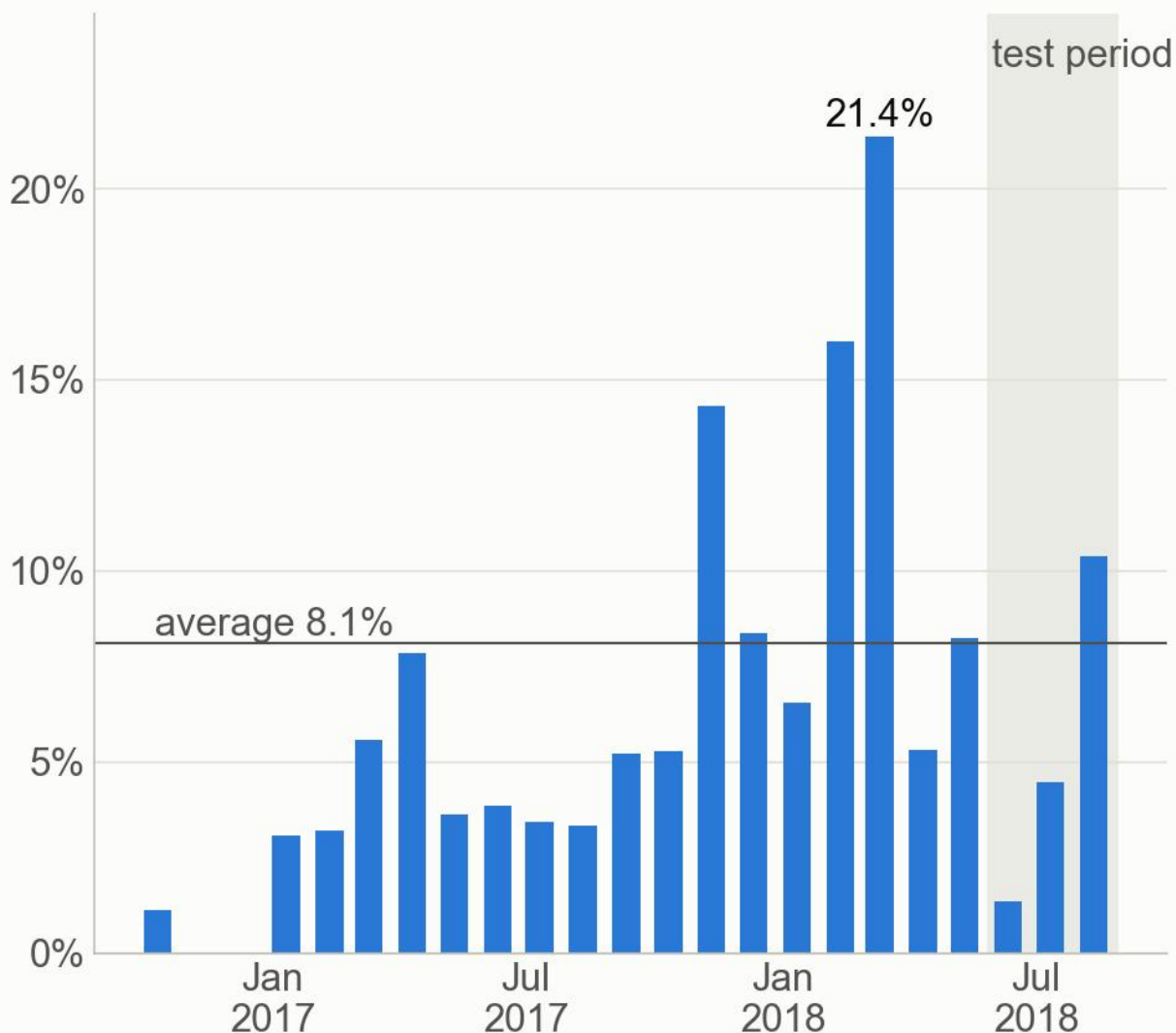
0.5 is not a neutral default

With ~9% late orders, few scores ever cross 0.5.
Precision, recall and F1 across every possible cut:



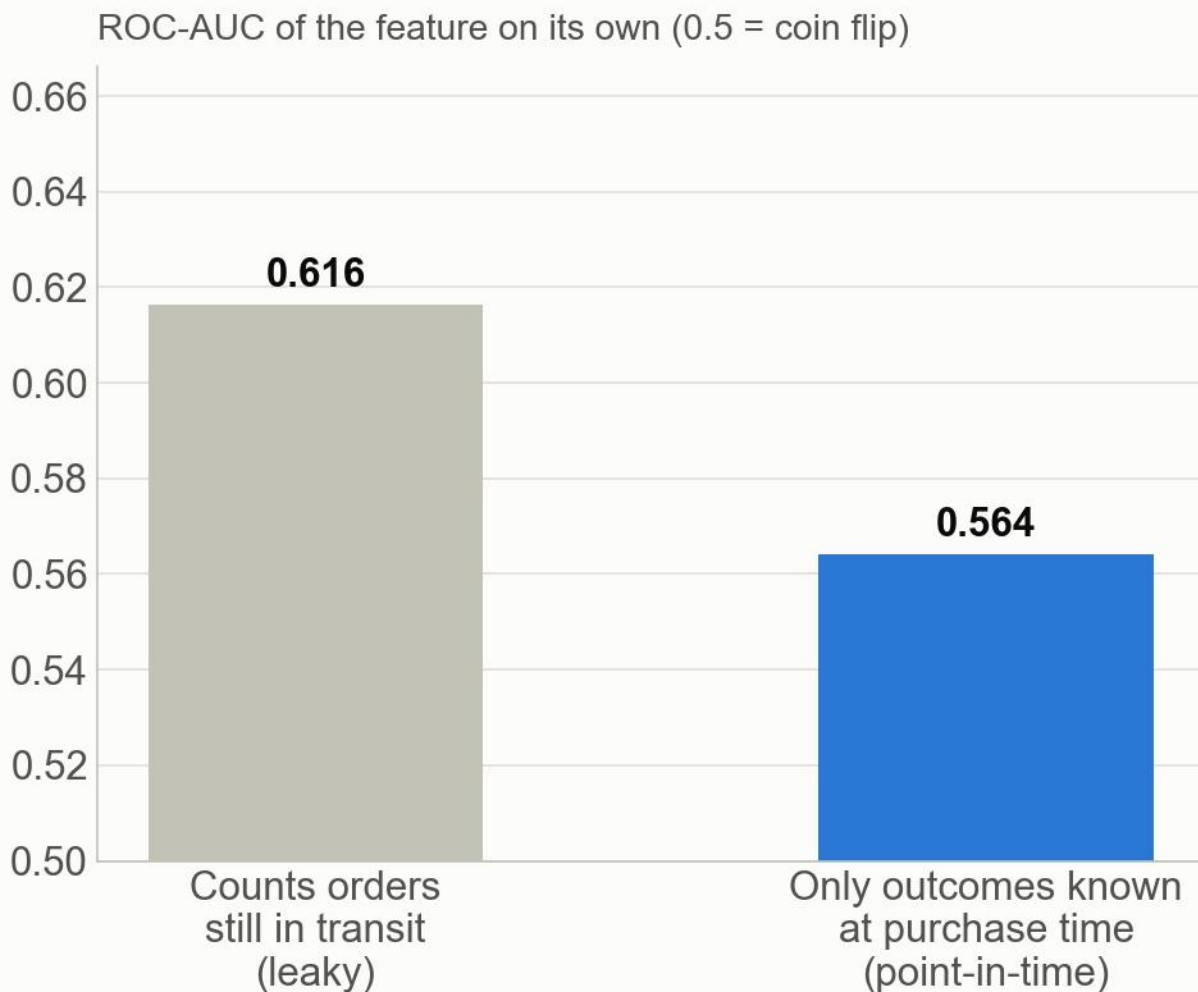
Delays are not stationary

The late-delivery rate swings from 1.1% to 21.4% month to month.
So every validation step here respects time.



The leak I found in my own feature

"Seller's historical late rate" counted orders still in transit at purchase time. Fixed, it looks weaker, and it's honest.



Tuning won the validation, lost the future

Optuna improved the temporal CV score, yet the untuned model ranked better on the unseen months.

